

CME396 Unit 3 – Cost Estimation Study Material

Oct 5, 2026 · @Legend

Exam pattern for Unit 3

Unit 3 carries **two 2-mark questions (Q5, Q6)** and **one 13-mark question (Q13 a/b)** in every paper. In 2024-25 the 13-mark question was a numerical (cost ladder or depreciation) in 3 of 4 papers, so practise the sums first.

Unit 3 syllabus: Costing vs estimation · importance and aims of estimation · elements of cost · ladder of cost · material and labour cost estimation · overheads and their allocation · depreciation · methods of costing.

Paper	Q5 (2 marks)	Q6 (2 marks)	Q13 (a)	Q13 (b)
Nov/Dec 2025	Estimation vs costing	Methods of overhead allocation	Cost ladder numerical	Depreciation numericals (reducing balance, machine hour)
Apr/May 2025	Purpose of cost estimation	One element of cost	Estimating labour and material cost	Overhead allocation and depreciation
Nov/Dec 2024	Overhead per machine (unit rate)	Man-hour and machine-hour rate	Cost estimation procedure for a new product	Appreciation and depreciation
Apr/May 2024	Importance of costing and estimation	Elements in allocation of overheads	Methods of costing (job, process, ABC)	Production cost in forging/welding/foundry

In Nov/Dec 2024, Q3 (administrative expenses) also came from this unit.

Formula sheet

Learn these formulas by heart: every Unit 3 numerical uses them.

1. Ladder of cost (each step adds one overhead):

- **Material consumed** = Opening stock + Purchases – Closing stock

- **Prime cost** = Direct material + Direct labour + Direct expenses
- **Factory (works) cost** = Prime cost + Factory overhead
- **Production (office) cost** = Factory cost + Administrative overhead
- **Total cost** = Production cost + Selling and distribution overhead
- **Selling price** = Total cost + Profit

2. Depreciation (C = initial cost, S = scrap value, n = life in years, H = total working hours over the life):

Method	Formula
Straight line	$D = (C - S) / n$ per year
Reducing balance	$r = 1 - (S / C)^{(1/n)}$; value after m years = $C (1 - r)^m$
Sinking fund	Annual deposit $A = (C - S) \times i / [(1 + i)^n - 1]$
Sum-of-years' digits	D in year $k = (C - S) \times (n - k + 1) / [n(n + 1)/2]$
Machine-hour basis	D per hour = $(C - S) / H$

3. Overhead allocation:

Method	Overhead charged to a job
% of direct material	$(\text{Total overhead} / \text{Total direct material cost}) \times 100 \times \text{job's material cost}$
% of direct labour	$(\text{Total overhead} / \text{Total direct labour cost}) \times 100 \times \text{job's labour cost}$
% of prime cost	$(\text{Total overhead} / \text{Total prime cost}) \times 100 \times \text{job's prime cost}$
Man-hour rate	$\text{Total overhead} / \text{Total man-hours}, \times \text{job's man-hours}$
Machine-hour rate	$\text{Total machine overhead} / \text{Total machine hours}, \times \text{job's machine hours}$
Unit rate	$\text{Total overhead} / \text{Number of units produced}$

4. Labour and material:

- **Material cost** = Weight (= Volume $\times$ Density) $\times$ Rate per kg, plus scrap and wastage allowance
- **Labour cost** = Time (set-up + machining + handling + allowances) $\times$ Wage rate per hour

Part A: 2-mark questions with answers

Write 3–4 crisp points per answer; a small table or formula earns the full 2 marks.

1. Why is cost estimation important? What is its purpose? (AM24, AM25)

Cost estimation is calculating the probable cost of a product **before** it is made. Its purposes are:

- to fix the selling price and quote for tenders
- to decide make-or-buy and choose the cheapest process
- to plan budgets and control costs
- to check whether the product is profitable before investing

2. Compare estimation and costing. (ND25)

Estimation	Costing
Probable cost, worked out before production	Actual cost, worked out during or after production
Done by the estimator (technical person)	Done by the cost accountant
Based on drawings, past data and judgement	Based on actual bills, wages and records
Used to fix price and quote	Used to control cost and check profit

3. List the elements of cost. (AM25)

- **Material:** direct and indirect
- **Labour:** direct and indirect
- **Expenses:** direct and indirect (overheads)

Overheads are split into factory, administrative, and selling & distribution overheads.

4. What are administrative expenses? (ND24)

The cost of running the office, not the shop floor. Examples: salaries of the manager and office staff, office rent, stationery, telephone, postage, legal and audit fees. They are added to the factory cost to get the production (office) cost.

5. List the methods of allocating overhead expenses. (AM24, ND25)

1. Percentage of direct material cost
2. Percentage of direct labour cost
3. Percentage of prime cost

4. Man-hour rate
5. Machine-hour rate
6. Combination of man-hour and machine-hour rate
7. Unit rate method

6. A firm makes 600 machines a year with total overheads of ₹1,80,000. Find the overhead per machine by the unit rate method. (ND24)

Overhead per machine = Total overhead / Number of units = $1,80,000 / 600 = ₹300$ per machine.

7. Define man-hour rate and machine-hour rate. (ND24)

- **Man-hour rate** = Total factory overhead / Total man-hours worked. Used where work is mainly manual.
- **Machine-hour rate** = Total cost of running a machine (depreciation, power, repairs, rent, supervision) / Machine hours worked. Used where work is mainly done on machines.

8. Define depreciation. List its methods.

Depreciation is the fall in an asset's value over time because of wear and tear, age or obsolescence. Methods: straight line, reducing balance, sinking fund, sum-of-years' digits, and machine-hour basis.

9. Define prime cost and factory cost.

- **Prime cost** = Direct material + Direct labour + Direct expenses
- **Factory cost** = Prime cost + Factory overhead

10. Differentiate direct and indirect material.

Direct material	Indirect material
Becomes part of the product	Helps production but is not part of the product
Can be traced to one job	Cannot be traced to one job
Example: steel bar for a shaft, castings	Example: lubricants, cotton waste, grinding wheels

Also revise: direct vs indirect labour (machine operator vs supervisor), and the meaning of "overhead" (all indirect costs: indirect material + indirect labour + indirect expenses).

Part B: 13-mark theory answers

Each answer below follows exam layout: definition → headings → points → example → conclusion. Draw the ladder-of-cost figure in any cost question; it earns easy marks.

Q1. Explain the step-by-step procedure for cost estimation of a new product. (ND24 13a)

Definition. Estimation is the technical process of finding the probable cost of a product before production starts, from its drawings and specifications.

Aims of estimation:

- fix the selling price and submit quotations or tenders
- decide make-or-buy
- choose the most economical method and machine
- prepare production budgets and check if the product can make a profit

Procedure:

1. **Study the drawing and specifications.** Note the material, dimensions, tolerances, finish and quantity.
2. **Prepare a bill of materials.** List every part, the bought-out items and the quantity of each.
3. **Estimate material cost.** Volume from the drawing → weight = volume × density → add allowances (machining, scrap, wastage) → weight × rate per kg.
4. **Decide the sequence of operations.** Turning, drilling, milling, welding, painting and so on, and the machine for each.
5. **Estimate time for each operation.** Set-up time + machining time + handling time + allowances (fatigue, personal needs, tool change).
6. **Estimate labour cost.** Time × wage rate per hour for each operation.
7. **Add direct expenses.** Special tools, jigs, patterns, design or testing charges, royalties.
8. **Find prime cost.** Material + labour + direct expenses.
9. **Add overheads.** Factory overhead → factory cost; administrative overhead → production cost; selling and distribution overhead → total cost.
10. **Add profit.** Total cost + profit = selling price.

Example (a cast-iron pulley): material 5 kg × ₹80 = ₹400; labour 2 hours × ₹150 = ₹300; direct expenses (pattern share) ₹50 → prime cost ₹750. With factory overhead 40% of labour (₹120), administrative ₹60 and selling ₹70, the total cost is ₹1,000. With 15% profit, the selling price is ₹1,150.

Conclusion. A correct estimate means competitive quotes, no losses on orders, and better cost control.

Q2. Explain the elements of cost and the ladder of cost.

Elements of cost:

1. Material

- Direct material: becomes part of the product (steel bar, castings, bought-out parts).
- Indirect material: helps production but cannot be traced to a job (oil, cotton waste, small tools).

2. Labour

- Direct labour: workers who directly change the product (turner, welder, fitter).
- Indirect labour: supervisors, storekeepers, inspectors, maintenance staff.

3. Expenses

- Direct expenses: chargeable to one job (special jigs, patterns, hire of a special machine, royalty).
- Indirect expenses (overheads): rent, power, lighting, depreciation, insurance.

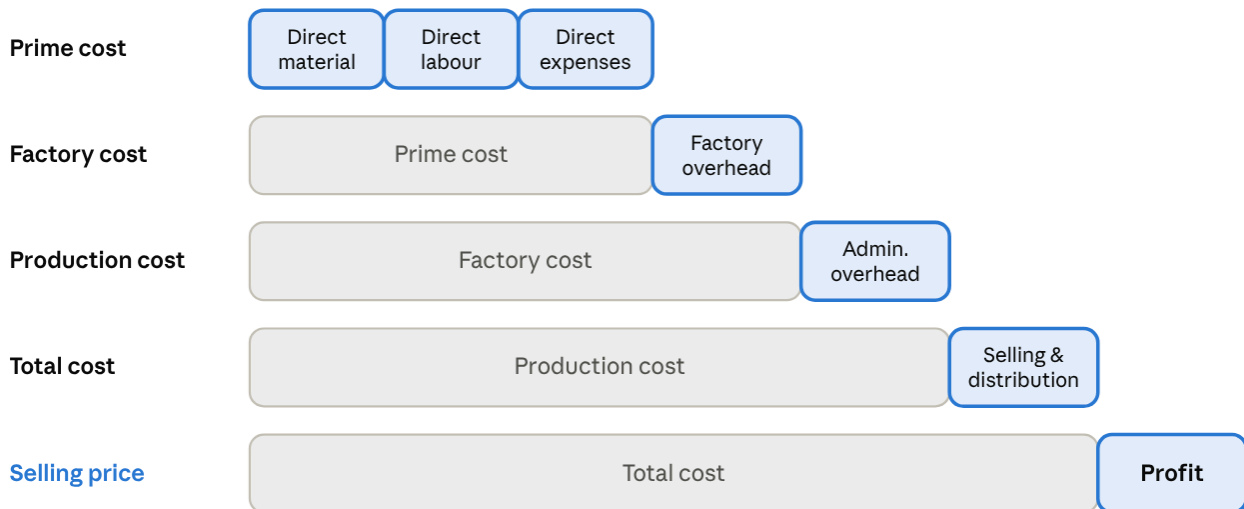
Overheads = indirect material + indirect labour + indirect expenses. They are grouped as factory, administrative, and selling & distribution overheads.

Ladder of cost: the cost is built step by step:

1. Direct material + direct labour + direct expenses = **Prime cost**
2. Prime cost + factory overhead = **Factory (works) cost**
3. Factory cost + administrative overhead = **Production (office) cost**
4. Production cost + selling and distribution overhead = **Total cost**
5. Total cost + profit = **Selling price**

The figure below shows the ladder; draw it in the exam.

Each step up the ladder adds one cost to reach the selling price



Ladder of cost · 5 steps from direct costs to selling price

Read each row as: grey block (the cost so far) + coloured block (the cost added) = the name on the left.

Q3. Explain the steps in estimating material cost and labour cost. (AM2513a)

A. Material cost estimation

1. Study the drawing and split the part into simple shapes (cylinders, prisms, cones).
2. Find the volume of each shape and add them (subtract holes).
3. Weight = volume × density (steel 7.86 g/cm³, cast iron 7.2 g/cm³, aluminium 2.7 g/cm³).
4. Add allowances: machining allowance, scrap or sprue loss, cutting (kerf) loss, scale loss in forging.
5. Material cost = gross weight × rate per kg, plus the cost of bought-out parts.
6. Subtract the scrap value if the scrap can be sold.

B. Labour cost estimation

1. List the operations from the process sheet.
2. Find the time for each: **set-up time** (preparing the machine), **operation time** (machining time from formulas or time study), **handling time** (loading, unloading, measuring).
3. Add allowances of about 10–20% for fatigue, personal needs, tool sharpening and delays.
4. Total time per piece = set-up time / batch size + operation time + handling time + allowances.
5. Labour cost = total time × wage rate per hour of that grade of worker.

Factors that affect labour cost: worker skill, machine type, batch size, wage rate, incentives and working conditions.

Example: a shaft needs 12 minutes turning + 3 minutes handling + 15% allowance = 17.25 minutes. At ₹120 per hour, the labour cost = $(17.25 / 60) \times 120 = \text{₹}34.50 \text{ per piece}$.

Q4. Explain overheads and the methods of allocating them. (AM24, AM25 13b, ND25)

Definition. Overheads are all indirect costs (indirect material, indirect labour and indirect expenses) that cannot be charged to one particular job. They must be shared out to every job fairly.

Classification:

Type	Examples
Factory (works) overhead	Factory rent, power, lighting, repairs, depreciation of machines, supervisor salary
Administrative (office) overhead	Office staff salaries, office rent, stationery, telephone, audit fees
Selling overhead	Advertising, salesmen's salaries and commission, showroom expenses
Distribution overhead	Packing, warehouse rent, transport and delivery vans

Methods of allocation (with formula and use):

1. **Percentage of direct material cost** = $(\text{Overhead} / \text{Direct material cost}) \times 100$. Used when material is the main cost.
2. **Percentage of direct labour cost** = $(\text{Overhead} / \text{Direct labour cost}) \times 100$. Simple; used when wage rates are uniform.
3. **Percentage of prime cost** = $(\text{Overhead} / \text{Prime cost}) \times 100$. Uses both material and labour.
4. **Man-hour rate** = $\text{Overhead} / \text{Total man-hours}$. Used in manual work; time-based, so more accurate.
5. **Machine-hour rate** = $\text{Machine running cost} / \text{Machine hours}$. The best method for machine shops.
6. **Combination of man-hour and machine-hour rates**. Used where both men and machines matter.
7. **Unit rate** = $\text{Overhead} / \text{Number of units}$. Used when all products are alike.

Example (percentage of labour): total overhead ₹50,000 and total direct labour ₹1,00,000 give 50%. A job with ₹2,000 labour is charged ₹1,000 overhead.

Conclusion. The machine-hour rate is the most accurate in modern machine shops because overhead is mostly caused by machines.

Q5. Explain depreciation, its causes and methods. Add a note on appreciation.
(ND24 13b, AM25 13b)

Definition. Depreciation is the permanent fall in the value of an asset (machine, building, tool) with time. It is charged as a cost so that money is set aside to replace the asset at the end of its life.

Causes:

- wear and tear from use
- age and decay (rust, corrosion)
- obsolescence: a newer, better machine is available
- accidents and breakdowns
- fall in demand for the product the machine makes

Methods:

1. **Straight line method.** Same amount every year: $D = (C - S) / n$. Simple, but ignores higher use in later years.
2. **Reducing balance method.** A fixed percentage of the remaining value each year: $r = 1 - (S / C)^{1/n}$. Gives higher depreciation in early years, when repairs are low.
3. **Sinking fund method.** A fixed amount is invested every year at interest i , so the fund grows to $(C - S)$ by the end of the life: $A = (C - S) \times i / [(1 + i)^n - 1]$.
4. **Sum-of-years' digits method.** Depreciation in year $k = (C - S) \times (\text{remaining life}) / (\text{sum of digits } 1 + 2 + \dots + n)$.
5. **Machine-hour basis.** $D \text{ per hour} = (C - S) / \text{total working hours}$. Fair for machines whose use varies.
6. **Annuity method.** Considers the interest lost on the money tied up in the machine.

Example (straight line): a machine costs ₹50,000 and has a scrap value of ₹5,000 after 10 years. $D = 45,000 / 10 = \text{₹4,500 per year}$.

Appreciation. Appreciation is the rise in an asset's value over time. Land, buildings in growing cities, and rare materials usually appreciate. Example: a factory plot bought for ₹10 lakh may be worth ₹15 lakh after 5 years. Appreciation is not charged to product cost, but it affects the value of company assets.

Q6. Explain the methods of costing. (AM24 13a)

1. **Job costing.** Cost is found separately for each job or order. Used in job shops, repair shops and special machines made to order. Each job has its own cost sheet.
2. **Batch costing.** A batch of identical items is treated as one job; cost per item = batch cost / number of items. Used for bolts, gears, medicines and toys.
3. **Process costing.** Used where the product passes through several processes in sequence (chemicals, cement, sugar, steel, paper). Cost is gathered for each process and passed to the next.
4. **Unit (single output) costing.** One standard product in large quantity (bricks, cement bags, mining). Cost per unit = total cost / units.
5. **Operating costing.** For services: transport (cost per km), power (cost per kWh), hospitals.
6. **Contract costing.** For large, long jobs such as buildings, bridges and ships; each contract is a cost unit.
7. **Activity-based costing (ABC).** Overheads are traced to activities (set-ups, inspection, material handling) and then charged to products by how much of each activity they use. More accurate where overheads are high and products vary.

Point	Job costing	Process costing	ABC
Product	Made to order, different each time	Continuous, identical	Many products with high overheads
Cost unit	One job	One process or period	One activity
Overhead	Broad rate	Per process	Per activity driver
Example	Repair of a machine	Cement plant	Car parts plant with many models

Solved numericals

These are the exact sums from past papers, worked step by step. Always write the formula first, then substitute; the formula alone earns step marks.

N1. Ladder of cost (ND25 13a)

Given (2023–24): opening stock of raw material ₹25,000, purchases ₹40,000, closing stock ₹15,000, direct labour ₹14,000, direct expenses ₹1,000, factory overhead ₹9,750, administrative ₹6,500, selling and distribution ₹3,250, 650 items, profit 10% of total cost.

Find the prime cost, factory cost, production cost, total cost and selling price per item.

Solution:

Step	Working	Total (₹)	Per item (₹)
Material consumed	25,000 + 40,000 – 15,000	50,000	76.92
Prime cost	50,000 + 14,000 + 1,000	65,000	100.00
Factory cost	65,000 + 9,750	74,750	115.00
Production cost	74,750 + 6,500	81,250	125.00
Total cost	81,250 + 3,250	84,500	130.00
Profit (10%)	0.10 × 130	8,450	13.00
Selling price	130 + 13	92,950	143.00

Answer: selling price = ₹143 per item.

N2. Reducing balance method (ND25 13b-i, 7 marks)

Given: cost C = ₹25,000, life n = 20 years, scrap S = ₹4,000. Find the depreciation fund after 3 years, the depreciation for the 3rd year, and the yearly rate.

Step 1: rate of depreciation

$$r = 1 - (S / C)^{(1/n)} = 1 - (4,000 / 25,000)^{(1/20)} = 1 - (0.16)^{0.05} = 1 - 0.9124 = \mathbf{0.0876}$$

So the value falls by **8.76% every year**.

Step 2: value year by year (value at end of year = previous value × 0.9124)

Year	Value at start (₹)	Depreciation at 8.76% (₹)	Value at end (₹)
1	25,000.00	2,188.90	22,811.10
2	22,811.10	1,997.24	20,813.86
3	20,813.86	1,822.33	18,991.53

Step 3: answers

- Depreciation fund after 3 years = 25,000 – 18,991.53 = ₹6,008.47
- Depreciation for the 3rd year = ₹1,822.33
- Value reduced every year by **8.76%**

Tip: to find $(0.16)^{0.05}$ without a calculator power key, use logs: $\log 0.16 = -0.7959$; $\times 0.05 = -0.0398$; antilog = 0.9124.

N3. Machine-hour basis (ND25 13b-ii, 6 marks)

Given: shaper cost $C = ₹85,000$, scrap $S = ₹8,500$ after $n = 8$ years, 3,330 working hours per year.

- Total working hours = $8 \times 3,330 = \mathbf{26,640 \text{ hours}}$
- Depreciation per hour = $(C - S) / \text{Total hours} = (85,000 - 8,500) / 26,640 = 76,500 / 26,640 = \mathbf{₹2.87 \text{ per hour}}$
- Annual depreciation = $2.8716 \times 3,330 = \mathbf{₹9,562.50 \text{ per year}}$

Answer: ₹2.87 per machine hour; ₹9,562.50 per year, which is $9,562.50 / 85,000 = \mathbf{11.25\%}$ of the cost each year.

N4. Unit rate method (ND24 Q5, 2 marks)

Overhead per machine = $1,80,000 / 600 = \mathbf{₹300 \text{ per machine.}}$

Practice problems (answers given; try them first)

P1. Straight line. A machine costs ₹1,20,000, has a scrap value of ₹20,000, and a life of 10 years. Find the yearly depreciation and the book value after 4 years.

- $D = (1,20,000 - 20,000) / 10 = \mathbf{₹10,000 \text{ per year}}$
- Book value after 4 years = $1,20,000 - 4 \times 10,000 = \mathbf{₹80,000}$

P2. Sum-of-years' digits. $C = ₹60,000$, $S = ₹6,000$, $n = 5$ years. Find the depreciation for years 1 to 3.

- Sum of digits = $1 + 2 + 3 + 4 + 5 = 15$; $C - S = 54,000$

Year	Fraction	Depreciation (₹)
1	5/15	18,000
2	4/15	14,400
3	3/15	10,800

P3. Ladder of cost. Direct material ₹30,000, direct labour ₹20,000, direct expenses ₹5,000. Factory overhead is 50% of direct labour, administrative overhead 10% of factory cost, selling overhead ₹5,000. 500 units are made and profit is 20% of total cost. Find the selling price per unit.

Step	Working	₹
Prime cost	$30,000 + 20,000 + 5,000$	55,000
Factory cost	$55,000 + 0.5 \times 20,000$	65,000
Production cost	$65,000 + 0.10 \times 65,000$	71,500
Total cost	$71,500 + 5,000$	76,500
Total cost per unit	$76,500 / 500$	153.00
Selling price per unit	153×1.20	183.60

P4. Machine-hour rate. A machine costs ₹2,00,000, scrap ₹20,000, life 10 years, 2,000 hours per year. It uses 5 kW of power at ₹8 per kWh. Repairs cost ₹10,000 per year and its share of rent is ₹6,000 per year. Find the machine-hour rate.

Item	Working	₹ per hour
Depreciation	$1,80,000 / (10 \times 2,000)$	9.00
Power	5×8	40.00
Repairs	$10,000 / 2,000$	5.00
Rent	$6,000 / 2,000$	3.00
Machine-hour rate		57.00

P5. Percentage of direct labour. The factory overhead for a year is ₹1,50,000 and the total direct labour is ₹3,00,000. A job uses ₹4,000 of material and ₹2,000 of labour. Find the factory cost of the job.

- Overhead rate = $1,50,000 / 3,00,000 \times 100 = 50\%$ of labour
- Overhead on the job = $0.50 \times 2,000 = ₹1,000$
- Factory cost = $4,000 + 2,000 + 1,000 = ₹7,000$

Last-minute revision checklist

Tick these off the night before; if all are done, Unit 3 is covered.

- ☐ I can write all 5 steps of the ladder of cost from memory and draw the figure
- ☐ I can solve N1 (ND25 cost ladder) without looking
- ☐ I know the reducing balance formula $r = 1 - (S/C)^{1/n}$ and can use logs to work it out

- ☐ I can solve N3 (machine-hour depreciation) and P4 (machine-hour rate)
- ☐ I can list the 7 methods of overhead allocation with their formulas
- ☐ I can list the 4 types of overhead with 2 examples each
- ☐ I can write the differences between estimation and costing (4 points)
- ☐ I can explain the 10-step cost estimation procedure with the pulley example
- ☐ I can explain the material and labour cost estimation steps
- ☐ I can write the causes and 5 methods of depreciation, plus appreciation
- ☐ I can compare job, process and activity-based costing in a table

Exam tips

- In numericals, show every step in a table; a wrong final number still earns most of the marks.
- Write units (₹, hours, %) on every answer.
- In theory answers, draw at least one table or the ladder figure; it makes the answer look complete.